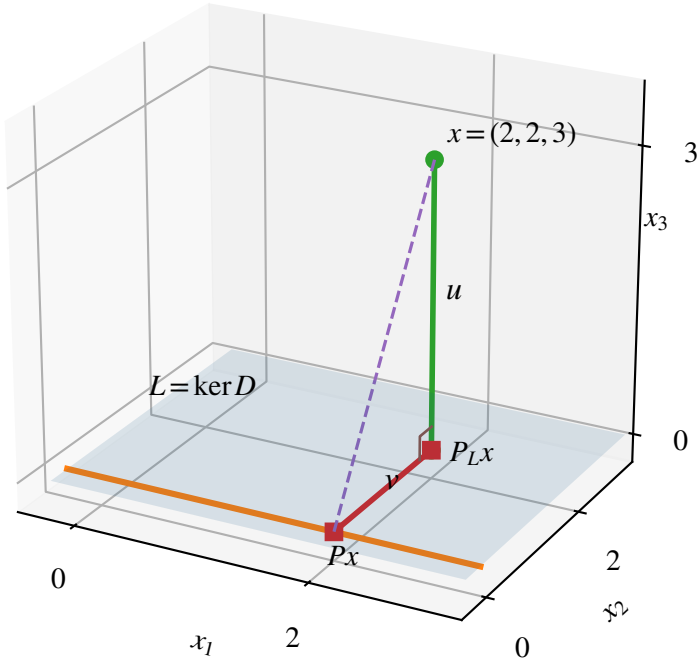

First correction

$u = (0, 0, 3)$

Second correction

$v = (0, 2, 0)$

$u \perp v$

$\|x - Px\|^2$

$= 3^2 + 2^2 = 13$